PRODUCT LCD MODULE
产品名称 ： 液晶显示模块
CMCS:
模块型号 ： JL-M101B196-P21WX-M402632
SUPPLIER SICHUAN JINGLONG PHOTOELECTRIC SCIENCE AND TECHNOLOGY CO.,LTD
供应商 ： 四川京龙光电科技有限公司
DATE
日期 ： 2026-04-23

SPECIFICATION
产品规格书

Version: V1.0
版本： V1.0
This module uses ROHS material
模块用环保材料

JL（京龙光电）		Customer
PREPARED BY 制定	黄贤明	
CHECKED BY 审核	陈志伸	
Quality Department 品质	李二涛	
Approved by 批准	彭武轮	Approved By ________________

四川京龙光电科技有限公司
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修改记录 **Revision Record**

版本号 version number	修改内容 Modifies the content	发起变更方 Initiating the changing party	日期 Date	修改人 Modifier
V1.0	首次发行 First issue	----	2026-04-23	黄贤明

目录 CONTENTS

1. General Specifications 基本规格

No.	Item 项目	Specification 规格	Unit 单位	Remark
1	LCD Size 液晶面板尺寸	10.1	Inch	
2	Panel Type 面板类型	IPS	-	-
3	Resolution 分辨率	800(H) X RGB X 1280(V)	Pixel	-
4	Display Mode 显示模式	Normally Black	-	-
5	Viewing Direction 使用视角	80/80/80/80 (U/D/L/R)	Deg.	-
6	Module Size 模组尺寸	143 x 228.6 x 2.6	mm	-
7	Panel Active Area 可视区域	135.36 X 216.576	mm	-
8	Pixel Pitch 像素尺寸	0.1692 X 0.1692	mm	-
9	Pixel Arrangement 像素排列	RGB - Stripe	-	-
10	Weight 重量	TBD	g	-
11	Back Light 背光	32 LED (4 串 8 并)	-	-
12	Interface 接口方式	MIPI	-	
13	Operating Temperature 工作温度	-10 ~ +50	℃	-
14	Storage Temperature 存储温度	-20 ~ +60	℃	-

2. Pin Assignments 接口定义

Pin no.	Symbol	Functional	Notes
1	NC	No connection	
2	VCC	Power supply +3.3V	
3	VCC	Power supply +3.3V	
4	GND	Ground	
5	RESET	Global reset	
6	NC	No connection	
7	GND	Ground	
8	D0N	MIPI data input 0-	
9	D0P	MIPI data input 0+	
10	GND	Ground	
11	D1N	MIPI data input 1 -	
12	D1P	MIPI data input 1 +	
13	GND	Ground	
14	CLKN	MIPI clock input -	
15	CLKP	MIPI clock input +	
16	GND	Ground	
17	D2N	MIPI data input 2-	
18	D2P	MIPI data input 2+	
19	GND	Ground	
20	D3N	MIPI data input 3-	
21	D3P	MIPI data input 3+	
22	GND	Ground	
23-24	NC	No connection	
25	GND	Ground	
26-29	NC	No connection	
30	GND	Ground	
31-32	LEDK	Back light -	
33-38	NC	No connection	
39-40	LEDA	Back light +	

3. Electrical Specification 电气特性

3.1 Absolute Maximum Ratings 极限参数

Item 项目	Symbol	Min.	Max.	Unit
Digital Supply Voltage 数字供电电压	VCC	1.8	4	V
Analog Power Supply Voltage 模拟供电电压	IOVCC	0.1	3.6	V

3.2 Typical Operation Conditions 典型工作条件

Item 项目	Symbol	Min.最小	Typ.典型	Max.最大	Unit
Analog Power Supply Voltage 模拟供电电压	VCC	2.8	3.0	3.4	V
Digital Supply Voltage 数字供电电压	IOVCC	1.65	1.8	3.3	V
Input High Voltage 输入高电平	VIH	0.7*IOVCC	-	IOVCC	V
Input Low Voltage 输入低电平	VIL	GND	-	0.3*IOVCC	V
Output High Voltage 输出高电平	VOH	0.8*IOVCC	-	IOVCC	V
Output Low Voltage 输出低电平	VOL	GND	-	0.2*IOVCC	V
Positive Power TFT 正电压	VGH	7	-	20	V
Negative Power TFT 负电压	VGL	-7	-	-15	V
Analog Power 伽玛校正电压	AVDD	4.5	-	6.3	V
external VCOM DC input	AVEE	-4.5	-	-6.3	V

3.3 Backlight Circuit Characteristics 背光功耗

Item	Symbol	Min.	Typ.	Max.	Unit	Remark
LED Current 背光电流	IB	--	160	--	mA	
LED Voltage 背光电压	Vf	11.2	12	12.8	V	
Power Consumption 功耗	PBL	--	1920	--	mW	
L/G Surface Luminance 模组表面亮度	Ls	280	300	--	cd/m^2	Note1
Surface brightness uniform 均匀度	Ld	75	80	--	%	Note2

Note1: Test condition is

(a) Center point on active area.

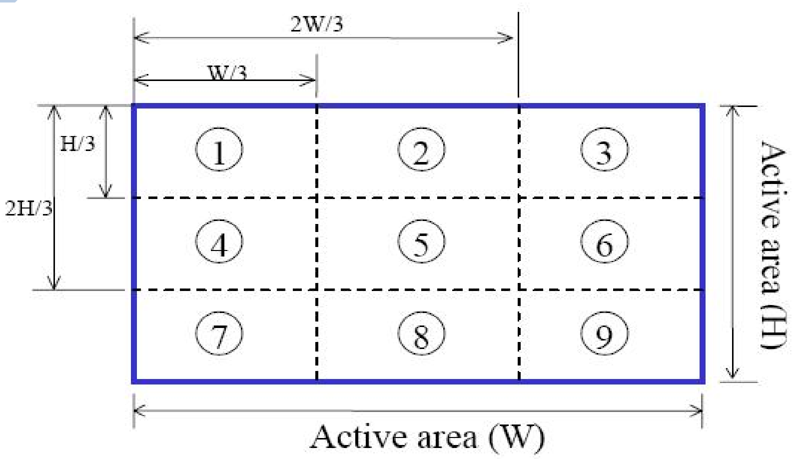
(b) Best Contrast.

Note2: Uniform measure condition:

(a)Measure 9 point. Measure location show below;

(b)Uniform=(Min. brightness /Max. brightness)*100%

(c)Best Contrast.



3.4 AC Characteristics

Reset input timing

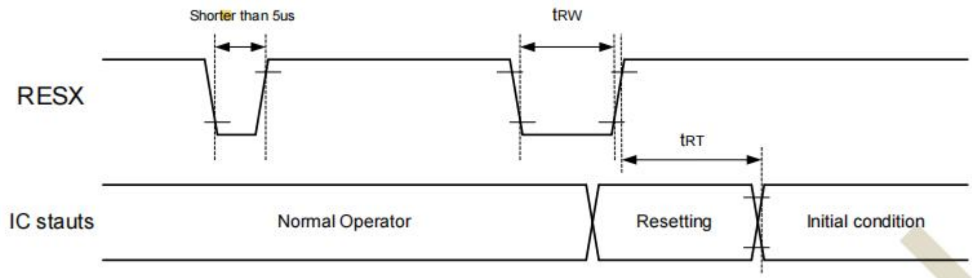


Figure 11.1: Reset input timings

Symbol	Parameter	Related pins	Min.	Max.	Unit
t_{RW}	Reset pulse width(2)	RESX	10	-	μs
t_{RT}	Reset complete time(3)	-	-	5 (Note 5)	ms
		-	-	120 (Note 6, 7)	ms

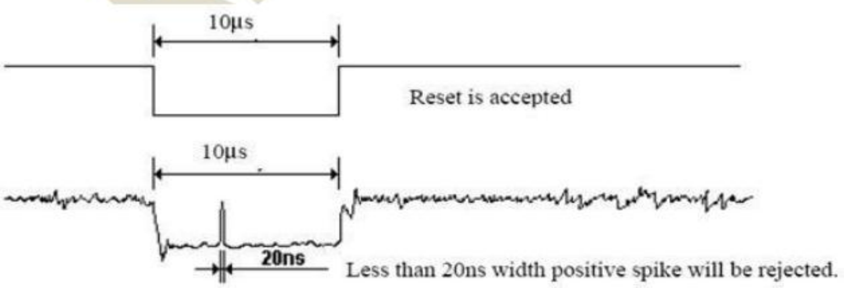
Note: (1) The reset complete time also required time for loading ID bytes from OTP to registers. This loading is done every time when there is HW reset cancel time (tRT) within 5 ms after a rising edge of RESX.

(2) Spike due to an electrostatic discharge on RESX line does not cause irregular system reset according to the table below.

RESX Pulse	Action
Shorter than 5 μs	Reset Rejected
Longer than 10 μs	Reset
Between 5 μs and 10 μs	Reset Start

(3) During the resetting period, the display will be blanked (The display is entering blanking sequence, which maximum time is 120 ms, when Reset Starts in Sleep Out –mode. The display remains the blank state in Sleep In –mode) and then returns to Default condition for H/W reset.

(4) Spike Rejection also applies during a valid reset pulse as shown below:



(5) When Reset is applied during Sleep In Mode.

(6) When Reset is applied during Sleep Out Mode.

(7) It is necessary to wait 5msec after releasing RESX before sending commands. Also Sleep Out command cannot be sent for 120msec.

(8) After Sleep Out Command, it is necessary to wait 120msec then send RESX.

3.5 DC Characteristics

Item	Symbol	Condition	Min.	Typ.	Max.	Unit
VCCD/IOVCC	V_{IN}	Interface Supply Voltage	1.65	-	3.6	
VCIP	V_{IN}	Logic Supply Voltage	2.5	-	6.0	
VCI	V_{IN}	Analog Supply Voltage	2.5	-	6.0	
VCCH	V_{IN}	High speed interface Supply Voltage	1.65	-	3.6	
Input high voltage	V_{IH}	VCCD/IOVCC= 1.65 ~ 3.3V VCIP= 2.5 ~ 3.3V VCI= 2.5 ~ 3.3V	0.7 VCCD/IOVCC	-	VCCD/IOVCC	V
Input low voltage	V_{IL}		0	-	0.3 VCCD/IOVCC	V
VPP	V_{IH} V_{IL}	VPP	8V	8.25V	8.5V	V
Output high voltage (SDO, LEDPWM)	V_{OH1}	I_{OH} = -1.0 mA	0.8 VCCD/IOVCC	-	VCCD/IOVCC	V
Output low voltage (SDO, LEDPWM)	V_{OL1}	VCCD/IOVCC= 1.65 ~ 2.4V I_{OL} = 1.0 mA	0	-	0.2 VCCD/IOVCC	V
Logic High level input current	I_{IH}	VSYNC, HSYNC	-	-	6	µA
		RESX, DCX_SCL, CSX, RDX, WRX_SCL	-	-	6	µA
	I_{IHD}	DB[23…0], SDI, DCX	-	-	6	µA
		DB[23…0]	-	-	6	µA
Logic Low level input current	I_{IL}	VSYNC, HSYNC	-6	-		µA
		RESX, DCX, CSX, RDX, WRX_SCL	-6	-		µA
	I_{ILD}	DB[23…0], SDI, DCX	-6	-		µA
		DB[23…0]	-6	-		µA
Current consumption standby mode (VCIP/VCI-VSSD)	$I_{ST(VDD)}$	VCIP/VCI=2.8V, VCCD/IOVCC=1.8V T_A =25°C	-	TBD	-	µA
Current consumption standby mode (VCCD/IOVCC– VSSD)	$I_{ST(VCCD/IOVCC)}$		-	TBD	-	µA
Current consumption during Deep-standby mode (VCIP/VCI-VSSD)	$I_{DP-ST(VDD)}$	VCIP/VCI=2.8V, VCCD/IOVCC=1.8V T_A =25°C	-	TBD	-	µA
Current consumption during Deep-standby mode (VCCD/IOVCC– VSSD)	$I_{DP-ST(VCCD/IOVCC)}$		-	TBD	-	µA

Note: 1. The VOTP pin is open on normal mode and in used while OTP programming condition.
2. The GRAM data is eliminated under the Deep standby mode.

3.6 Power on/off Sequence 上电时序

3.6.1 General

During power off, if the display module is in the SLPOUT mode, VCI, VCIP and VCCD/IOVCC must be powered down minimum 120msec after RESX has been released.

During power off, if the display module is in the SLPIN mode, VCI, VCIP and VCCD/IOVCC can be powered down minimum 0msec after RESX has been released.

There will be no damage to the display module if the power sequences are not met. There will be no abnormal visible effects on the display panel during the Power On/Off Sequences.

There will be no abnormal visible effects on the display panel between end of Power On Sequence and before receiving SLPOUT command. Also between receiving SLPOUT command and Power Off Sequence.

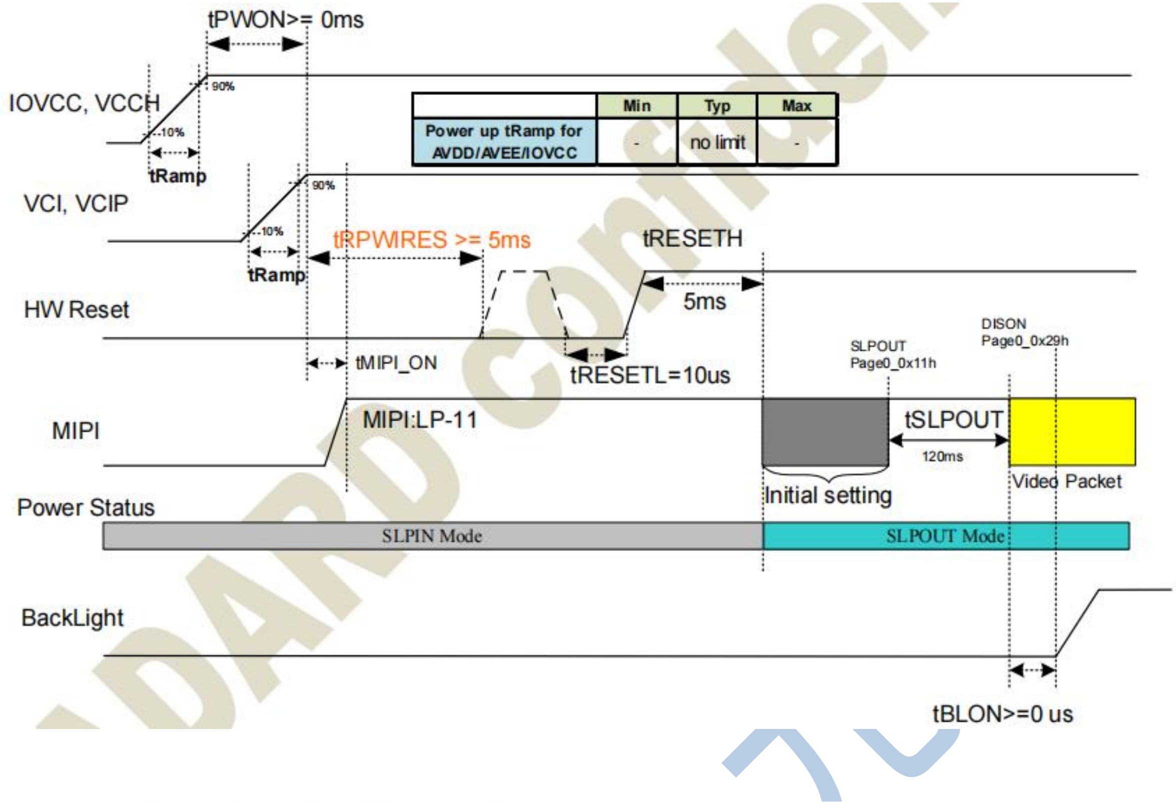
If RESX line is not held stable by host during Power On Sequence as defined in Sections 9.5.2, then it will be necessary to apply a Hardware Reset (RESX) after Host Power On Sequence is complete to ensure correct operation. Otherwise function is not guaranteed.

There is not a limit for Rise/Fall time on VCI, VCIP and VCCD/IOVCC.

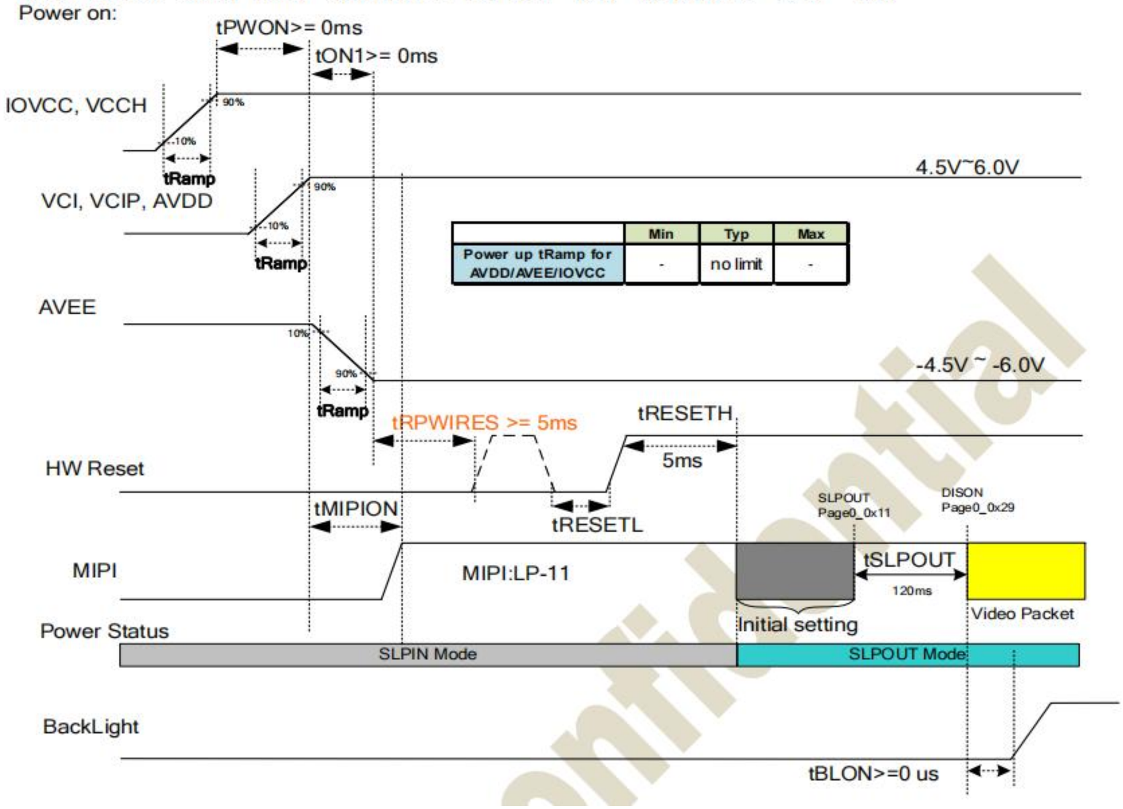
3.6.2 Power on sequence for differential power mode

Symbol	Min	Typ	Max	Unit	Remark
tRamp	-	no limit	-	us	
tPWON	0	-	-	ms	
tON1	0	-	-	ms	
tMIPI-ON	0	-	tRPWIRES	ms	
tRPWIRES	5	-	-	ms	
tRESETL	10	-	-	us	
tRESETH	5	-	-	ms	
tSLPOUT	120	-	-	ms	
tBLON	0	-	-	ms	

BOOSTM[1:0]=10 (Internal DC/DC power mode : Charge Pump, FP7721)
VCCD=IOVCC=VCCH=1.65V ~ 3.6V, VCI=VCIP=2.5V ~ 4.8V.



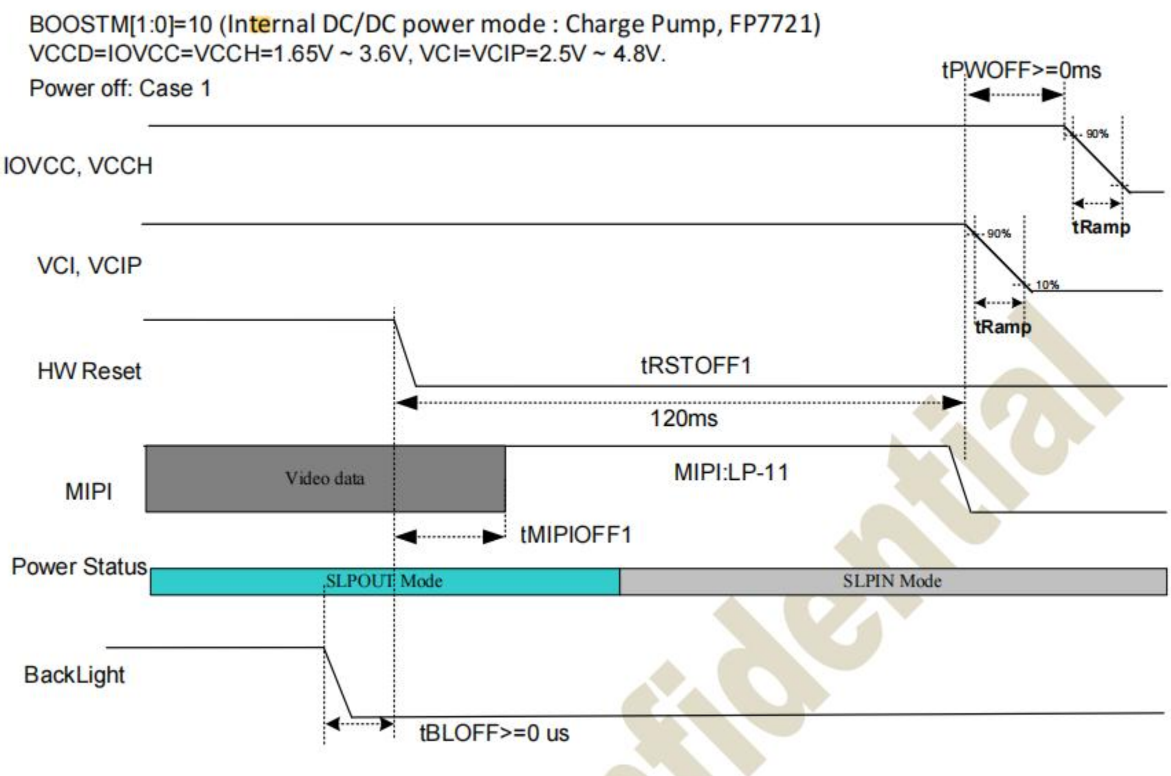
BOOSTM[1:0]=01/11 (External AVDD/AVEE Power)
VCCD=IOVCC=VCCH=1.65V ~ 3.6V, AVDD=VCI=VCIP=4.5V ~ 6.0V, AVEE=-4.5V ~ -6.0V
Power on:



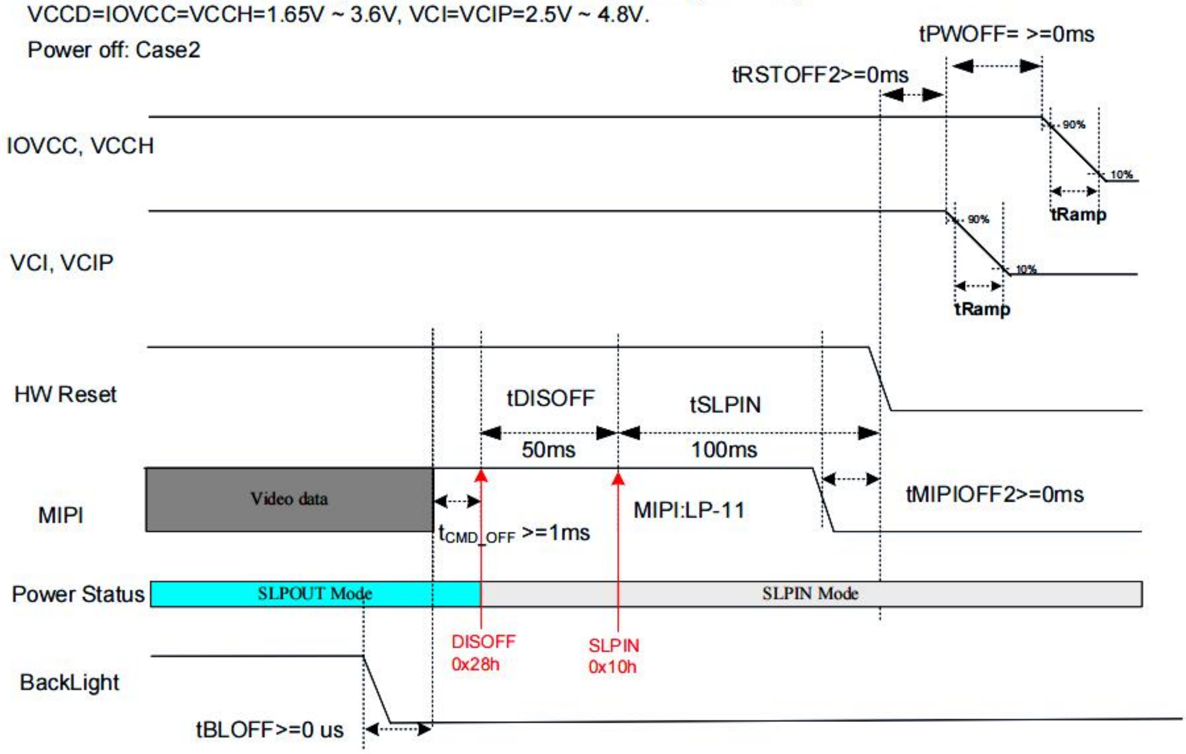
3.6.3 Power off sequence for differential power mode

For the power off case2, DISOFF command and tDISOFF are optional. That means t_{CMD_OFF} could be followed by the SLPIN command and tSLPIN , without DISOFF command and tDISOFF.

Symbol	Min	Typ	Max	Unit	Remark
tRamp	-	no limit	-	us	
tPWOFF	0	-	-	ms	
tPWOFF1	0	-	-	ms	
tPWOFF2	0	-	-	ms	
tMIPIOFF1	0	-	-	ms	power off case 1
tRSTOFF1	120	-	-	ms	power off case 1
tMIPIOFF2	0	-	-	ms	power off case 2
tRSTOFF2	0	-	-	ms	power off case 2
tCMD_OFF	1	-	-	ms	power off case 2
tDISOFF	50	-	-	ms	power off case 2
tSLPIN	100	-	-	ms	power off case 2
tBLOFF	0	-	-	ms	

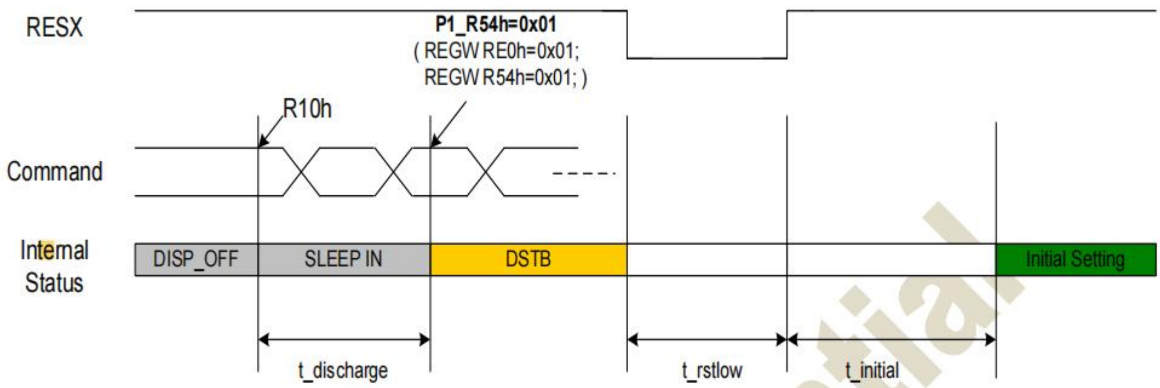


BOOSTM[1:0]=10 (Internal DC/DC power mode : Charge Pump, FP7721)
VCCD=IOVCC=VCCH=1.65V ~ 3.6V, VCI=VCIP=2.5V ~ 4.8V.
Power off: Case2



3.6.4 Deep standby flow

The figure below illustrates deep standby flow, :



Signal	Symbol	Parameter	MIN	TYP	MAX	Unit	Description
RESX	$t_{discharge}$	Sleep in into DSTB delay time	100	-	-	ms	
	t_{rstlow}	Reset low pulse	5	-	-	ms	
	$t_{initial}$	Reset high to initial setting delay time	10	-	-	ms	

Note 1) $t_{\_discharge}$ suggested delay time over 100ms.

Note 2) $t_{\_initial}$ suggested delay time over 10ms..

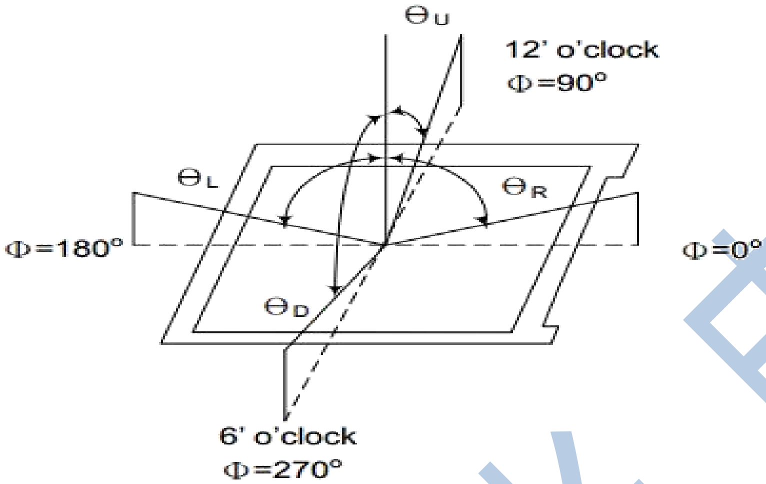
4. Optical Specification 光学参数

4.1 LCD Optical Characteristics 液晶模组光学特性

Parameter		symbol	condition	min.	typ.	max.	unit	remark
viewing angle range	horizontal	θR	CR>10	75	80	–	Deg.	Note 1
		θL		75	80	–	Deg.	
	vertical	θU		75	80	–	Deg.	
		θD		75	80	–	Deg.	
Luminance Contrast ratio		CR	θ=0°	800	1000	–		Note 2
Cell Transmittance		Tr		–	5.8	–	%	
white chromaticity		Xw		TYP. -0.03	0.283	TYP. +0.03		
		Yw			0.307			
Reproduction of color (C light)	red	Rx			0.62			
		Ry			0.367			
	green	Gx			0.339			
		Gy			0.597			
	blue	Bx			0.154			
		By			0.105			
Color gamut (clight)				52	56		%	
Response Time (Rising + Falling)		Trt	Ta=25℃	–	30		ms	

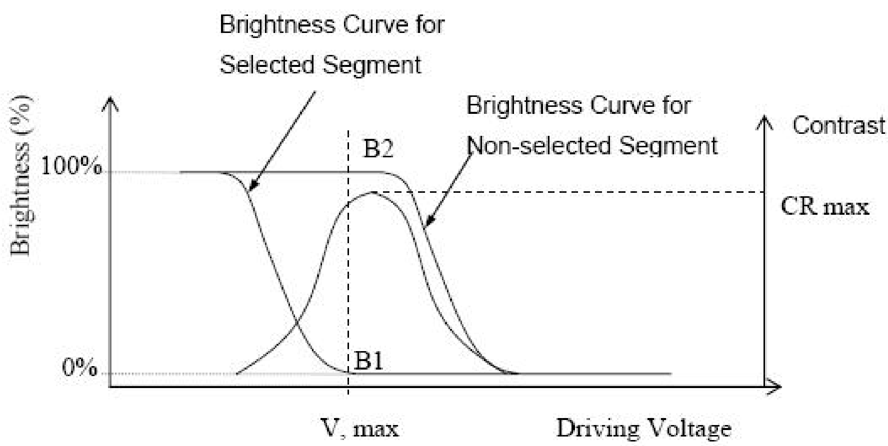
4.2 Measurement system 测量系统

Note 1： LCM Viewing Angle

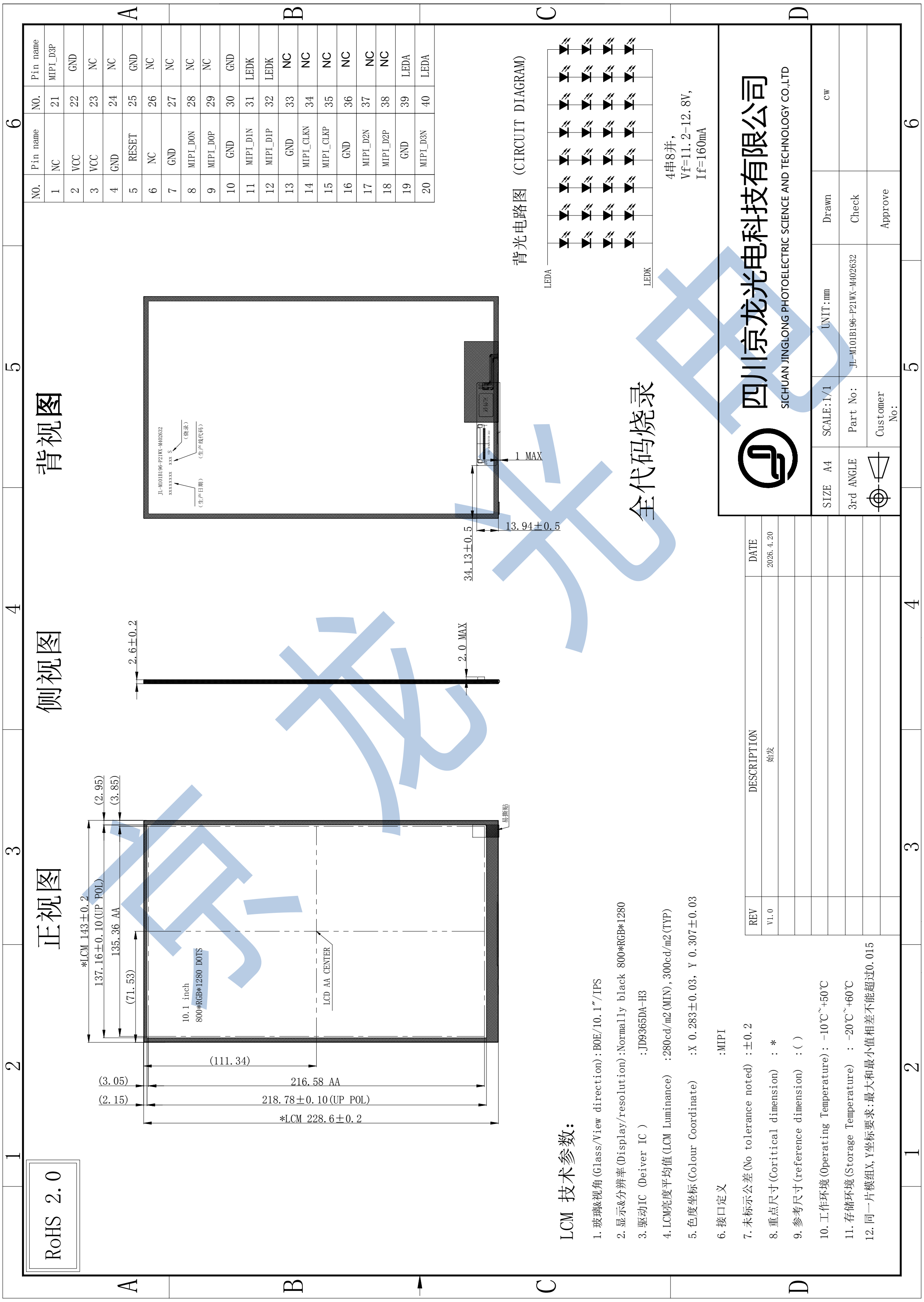


Note 2： Definition of Contrast Ratio(CR) : measured at the center point of panel

$$CR = \frac{\text{Brightness of Non-selected Segment (B2)}}{\text{Brightness of Selected Segment (B1)}}$$



5. Module Drawing 模组图



6. Reliability Test Items 可靠性测试项目

Test Item 测试项目	Test Condition 测试条件	Test result determinant gist 实验结果判定
High temperature storage 高温存储	60±3℃，96H；	Inspection after 2~4hours storage at room temperature, the sample shall be free from defects: 试验结束后,已测试的 LCD 样品必须在室内正常温湿度环境下放置 2~4 个小时以上才能进行功能和外观检查，样品不允许有以下缺陷： 1.Air bubble in the LCD;模块中有气泡； 2.Non-display; 不显示； 3.Glass crack; 玻璃破碎； 4. The electrical characteristics requirements shall be satisfied. 需要满足模块电气性能。
Low temperature storage 低温存储	-20±3℃，96H；	
High temperature operation 高温运行测试	50±3℃，96H；	
Low temperature operation 低温运行测试	-10±3℃，96H；	
High temperature /humidity 高温高湿	50℃±3℃,90%±3%RH，24H；	
Thermal Shock 冷热冲击	-20℃/0.5h~+60℃/0.5h for a total 24 cycles；	
Vibration Test 振动测试	Frequency 10Hz~55Hz~10Hz Amplitude：1.5mm, X，Y，Z direction for total 1H； (Packing condition)	
ESD test 静电测试	±4KV, Human Body Mode, 150pF/330Ω；±8KV, Air Mode, 150pF/330Ω；	

Remark: 注意：

1. The test samples should be applied to only one test item.
每个被测试的模块只能用于其中的一个测试项目。
2. Sample size for each test item is 2pcs.
每个测试项目的样品数量为 2 片。
3. Failure Judgment Criterion: Basic Specification, Electrical Characteristic, Mechanical Characteristic Optical Characteristic.故障判断标准:基本规格,电气特性,机械特性,光电特性。

7.Storage Method 存储方法

1.Store in an ambient temperature of 23°C±5°C, and in a relative humidity of 55%±15%. Don't exceed 12 months and expose to sunlight or fluorescent light.

存储环境温度为 23±5°C，相对湿度为 55%±15%，存储不能超过 12 个月，不要长时间暴晒。

2. Store in a clean environment, free from dust, active gas, and solvent.

存储在一个干净的环境，不受灰尘，活性气体和溶剂污染。

3. Store in antistatic container.

存储在防静电环境。

8.Announcements 注意事项

1.Do not attempt to disassemble or process the LCD module.

请勿拆卸液晶显示模块。

2.Do not make extra holes on the printed circuit board, modify its shape or change the positions of components to be attached。

不要在印制电路板上钻额外的孔，修改形状或更改印制线路板上元件的位置。

3.Except for soldering the interface, do not make any alterations or modifications with a soldering iron; Ensure welding temperature at 320 ° C to 350 ° C, the welding time control within the 10 s, welding note don't stay too long in the same place to avoid scald FPC.

除焊接接口外，不要用烙铁做任何更改；焊接温度保证在 320°C-350°C，焊接时间控制在 10S 以内，焊接时注意不要在同一处停留时间太久以免烫伤 FPC。

4. Other matters in not clear before use, please contact our staff to guide.

其他事项在不清楚使用之前，请联系我司人员指导进行。

9.供应商联系信息/Supplier contact information

Factory1: No. 305, Section 3, West Section of North Changjiang Road, Shaping Sub-district, Lingang Economic Development Zone, Yibin City, Sichuan；

生产基地 1：四川省宜宾市临港经开区沙坪街道长江北路西段附三段 305 号；

Shenzhen Office: 2 / F, building 5, Huafeng Industrial Zone, Fengtang Avenue, Fuyong Town, Bao'an District, Shenzhen

深圳办事处：深圳市宝安区福永镇凤塘大道华丰工业区 5 栋 2 楼；

-END-